

Muhammad Anas Khan

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Objective:

Final-year CS student with hands-on full-stack development, DevOps and AI/ML experience. Seeking a software engineering role to build scalable web/mobile and AI systems.

Education:

- **Bachelor of Computer Science**, FAST - National University of Computer and Emerging Sciences KHI (2022-Present)
CGPA: 3.13

Experience:

Web Development Intern (MERN Stack) – OutSecure, Rawalpindi

Jul 2025 – Sep 2025

- Completed a structured remote internship program focused on React app development with MERN stack.
- Gained exposure to collaborative workflows and industry-standard tools.

DevOps Intern – QBS co., Karachi

December 2025 – Present

- Managing CI/CD, containerization, and server infrastructure, cloud-native and on-prem deployments.
- Developing gRPC APIs, monitoring logs, and integrating MLOps workflows, ensuring scalable & secure backend systems

Skills:

Languages & Scripting: Python, C++, JavaScript, Go, Bash, Kotlin

Web & Backend: React, MERN Stack, FastAPI, REST, gRPC APIs

Databases & Cloud: PostgreSQL, MongoDB, MySQL, AWS, Github

DevOps: Docker, Kubernetes (K8s), Bash, CI/CD, Linux, Jaeger Tracing, Grafana, Loki, Prometheus, Monitoring, Jenkins, Git

ML/AI: NumPy, Pandas, Scikit-learn, PyTorch, Keras

Soft Skills: Fast learner • Communication • Teamwork

Projects:

- **Final Year Project – Offline Payment System for Pakistan** (Ongoing, 2025–2026)
 - Developing an Android application using **Kotlin** to support **offline peer-to-peer payments** through QR-based transaction exchange.
 - Building a **FastAPI backend** to handle offline transaction recording, deferred synchronization, and basic double-spending checks.
 - Implementing secure transaction handling using **asymmetric cryptography**, nonces, and local secure storage, with future integration planned for Android Keystore.
 - **Tech Stack:** Kotlin, Android (Jetpack Compose), Room (SQLite), Python, FastAPI, PostgreSQL, REST APIs, Cryptography, QR codes, Cloud deployment
 - GitHub: [Offlink-Android-App](#)

- **gRPC Log Collector – Distributed Tracing Service** (*Proprietary – QBS Co.*) **(2026)**
 - Built a production-ready log collection service in **Go** with **gRPC** (unary + client-streaming RPCs), implementing a worker pool with panic isolation and bounded queue for non-blocking, resilient log ingestion across microservices.
 - Integrated **OpenTelemetry SDK** with OTLP/gRPC export to **Jaeger**, propagating nested distributed traces across a 4-service chain (frontend → backend → Service1 → Service2) with full span attribute and error status tracking.
 - Deployed via **Docker Compose** with per-service Prometheus metrics endpoints and **Slack webhook** alerts for real-time error and queue saturation notifications.
 - **Tech Stack:** Go, gRPC, Protocol Buffers, OpenTelemetry, Jaeger, Docker, Prometheus
- **Centralized Monitoring & Log Aggregation** (*Proprietary – QBS Co.*) **(2026)**
 - Built an on-prem centralized logging & monitoring system using Kubernetes (MicroK8s), Grafana, Loki, and Fluent Bit.
 - Deployed Grafana & Loki on Kubernetes, exposed Loki via NodePort, and ingested logs from remote Linux servers.
 - Created Grafana dashboards and LogQL queries for real-time log analysis, troubleshooting, and operational visibility.
- **Deep Learning Model – Bacterial Colony Classification** **(2025)**
 - Developed an image classification model using EfficientNetB0 transfer learning to classify 33 bacterial colony classes from a small, imbalanced dataset.
 - Implemented data preprocessing and augmentation (image cleaning, resizing, normalization) and applied fine-tuning of deep CNN layers to improve generalization.
 - Trained and evaluated the model across multiple phases, analyzing accuracy/loss trends and overfitting behavior to guide architectural and training decisions.
 - Designed a Streamlit-based inference workflow (prototype) for image upload and prediction as part of model deployment exploration.
 - **Tech Stack:** Python, TensorFlow, Keras, EfficientNetB0, Streamlit, Google Colab, Kaggle API
 - GitHub: [Bacteria-Classifier](#)
- **Backend Development – Multi-Store Inventory Management System** **(2025)**
 - Built a scalable RESTful backend supporting 500–1,000+ stores, enabling concurrent inventory operations with strict per-store data isolation.
 - Implemented JWT-based authentication and role-based authorization, enforcing secure access via middleware and preventing unauthorized cross-store actions.
 - Designed an optimized PostgreSQL relational schema with event-driven asynchronous audit logging and Redis caching to ensure data integrity, traceability, and high-performance reads.
 - Hardened APIs with rate limiting, bcrypt password hashing, negative stock prevention, and SQL injection-safe queries.
 - **Tech Stack:** Node.js, Express.js, PostgreSQL, Redis, JWT, EventEmitter
 - GitHub: [Kiryana Store Web Backend](#)

Interests:

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| • Web development | • Machine Learning/AI |
| • Android Development | • Cyber Security |
| • DevOps | • Cross Platform Development |